

Constructing An Environment Agnostic Tuberculosis Predictor From Ziehl-Neelsen Stained Sputum Smears Using A CNN

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Abstract

Tuberculosis is one of the world's most infectious and deadly diseases. Tuberculosis affects about 25% of the world's population and is responsible for roughly 1.2 million deaths each year. In this work I investigate using a CNN model to assist health officials in the diagnosis of Tuberculosis via likelihood classification of sputum slides stained with the Ziehl-Neelsen staining technique. Through my methods I achieved a 0.9198 validation accuracy along with a 0.2097 binary cross-entropy validation loss and showed that the model I trained is likely generalizable to other datasets and microscopy environments.

1. Introduction

According to the World Health Organization (WHO) approximately a quarter of the world's population is currently infected with Tuberculosis (Bai & Ameyaw, 2024). In addition, the disease has been shown to be much more likely to have extreme adverse health impacts and be much harder to cure for those living in poverty. The best way to prevent the spread of Tuberculosis in these places is being able to diagnose the disease early, however most diagnostic tools are very expensive and the people living in these impoverished communities do not have access to them.

The current most widespread diagnostic tool for Tuberculosis in impoverished areas is manual Ziehl Neelsen stained sputum smear microscopy. The Ziehl-Neelsen stain is used to detect Tuberculosis and other acid-fast bacteria which have a waxy lipid-rich outer layer which causes them to be resistant to conventional staining techniques. The microscopy technique requires a trained technician to examine a sputum smear from a patient under a microscope and look for Tuberculosis bacteria manually. This method is time-consuming and has the potential to overlook between 33%-55% of cases (Veropoulos et al., 1998). Further, "Specialists are required to analyze between 20 and 80 fields on a single slide, a process that can take approximately 40 minutes to 3 hours" (Tamura et al., 2024).

Because of the low detection rate of the manual method, I have created a CNN image classification model which can be used to assist lab technicians by cropping a microscope viewfield into multiple (224×224) pixel image sub-croppings and sort the crops by their likelihood of containing Tuberculosis Bacilli. This proposed method has the potential to greatly speed up the lengthy and error prone process of manually detecting Tuberculosis Bacilli.

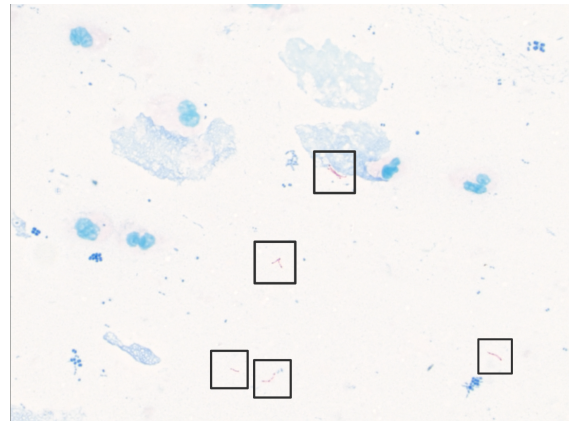


Figure 1. Example of (224×224) px croppings from ODS3, used for training the model.

2. Related Work

A description of similar work in the area of using machine learning models for Tuberculosis Bacilli detection on Ziehl-Neelsen sputum smears:

Automatic bright-field smear microscopy for diagnosis of pulmonary Tuberculosis: The authors of this paper also used a CNN to detect Tuberculosis (Serrão et al., 2024). They evaluated multiple architectures and even used an encoder-decoder architecture as opposed to a generic CNN architecture, finding it to have better results.

While the authors of this paper had good success in training their model, I believe that my proposed methods are improvements in that their crop size was much larger than

mine at (400×400) rather than (224×224) . A crop size of (400×400) not only requires a larger model to process and therefore greater computing power, but also has a greater potential of using extraneous domain specific artifacts in its reasoning.

Further, the authors of this paper only used one dataset for their input images, choosing to strengthen their model through augmentation, while I plan to use multiple distinct datasets in addition to augmenting my training images.

Pulmonary Tuberculosis Diagnosis Using an Intelligent Microscopy Scanner and Image Recognition Model for Improved Acid-Fast Bacilli Detection in Smears: The authors of this paper were the ones who constructed the dataset the Wellgen Medical Subset was based off of (Chen et al., 2024). This study constructed a CNN model using 100,000 sputum slides. Due to the large amount of input images, their model was able to perform very well. However, due to it being trained on macroscopic images, rather than smaller crops like my model was, it is potentially possible that their method generalizes worse than mine because as there is a greater chance that it uses domain specific features in its decision process, so it might struggle when ran in foreign lab environments. Further, a CNN model trained to handle (2448×2048) input images would likely be very large and computationally expensive to run, and therefore not useful in underfunded laboratories.

My method can potentially be an improvement on their method because, rather than use the entire slide as input, I extract crops which are easier for the CNN model to pull general and meaningful features out of and make inferences on. In addition, my model is likely much smaller and easier to run than theirs, requires less training data to get adequate results, and has a lesser chance that foreign objects and image features are used in its decisions.

Automated Mycobacterium Tuberculosis Detection in Multivariant Digitized Ziehl-Neelsen Staining Using Faster R-CNN Method: This paper outlines training a Faster R-CNN model on the TBDB (Trilaksana et al., 2020) dataset. Their methods retained the (612×480) image size for training their R-CNN and in addition augmented saturation, color, brightness, stretched the images, and added blur.

There are definitely merits to the augmentations done by this team, however I believe that my model is an improvement over their methods because my crop size is smaller, meaning my model is able to pick up on features of the small Tuberculosis Bacilli quicker than theirs, while requiring less computational cost to run. Further, I use more diverse datasets, meaning my model is likely more robust to variations in lab equipment and practices and would therefore perform better in the real world.

3. Methodology

The main approach I used to train the CNN is to crop the dataset images from their starting resolution to a resolution of (224×224) pixels around labeled Tuberculosis Bacilli. I then tested my trained model on a dataset of slides without labels for the locations of the Tuberculosis Bacilli and made use of a custom crop viewer constructed to get an understanding of the real world performance of my model.

I generated the (224×224) crops from the training datasets using the following methods:

ODS3 dataset: The Original Dataset 3 (ODS3) (Costa et al., 2025) is a dataset which was compiled using 3,280 extended depth focus bitmap images along with comma separated value (CSV) files labeling the location of Tuberculosis Bacilli present on the image. The CSV uses “b” or “bc” to label bacillus or bacillus cluster in the related bitmap image, along with coordinates for where they can be found on the image.

I cropped the bitmap format images from the ODS3 into (224×224) pixel crops surrounding the labeled bacillus. This allowed the CNN model a closer resolution to better capture the specific features of the Tuberculosis bacteria. Creating these crops from the labeled Tuberculosis Bacilli resulted in 10,897 images containing Tuberculosis Bacilli, labeled positive. I then compiled a similar number of (224×224) image crops without bacillus by cropping images from slides without corresponding annotation files, these crops were labeled negative. I took 4 crops per image this way which resulted in 9,726 negative crops, roughly equal to the number of positive crops.

TBDB dataset: The Tuberculosis Database (TBDB) (Trilaksana et al., 2020) contains 600 (612×480) resolution images of Tuberculosis sputum smears along with XML files containing bounding boxes detailing the location of the Tuberculosis Bacilli and debris on the image.

To make this data compatible with my model architecture, I cropped out (224×224) segments from these images where there was a Tuberculosis Bacillus labeled as present. This resulted in 1,482 positive image crops. In addition, to generate negative class images which can help to train my model against false positives, I made crops around the annotation locations which were labeled as debris and labeled the 1,585 resulting crops negative.

As the crops generated from this dataset are slightly larger than the ones generated from the ODS3 dataset, it is useful in training the model to detect Tuberculosis Bacilli of more varying sizes.

Testing via Wellgen Medical Subset: The Wellgen Medical subset (Lin & Medical, 2024) is a 1000 image subset of a

large dataset of Ziehl-Neelsen sputum smear slides divided into (2448×2048) regions, released by Wellgen Medical.

The slides in this dataset are labeled positive and negative, but none of the positive slides have bounds around the locations of the Bacilli. As a result, I cannot use this dataset to train the CNN model. However, I found this dataset useful in getting an idea of how well my model generalizes to data it hasn't been trained on.

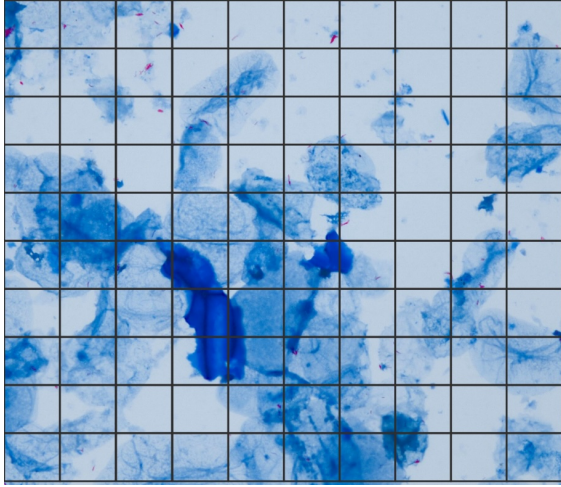


Figure 2. Example of the 90 testing (224×224) px croppings taken from a positive slide with no location labels.

Specifically, to test models using the images in this dataset I separate the (2448×2048) images into 90 (224×224) crops (See Figure 2). I then have the model being tested predict the probability that Tuberculosis Bacilli are present in each of the crops and if the probability is greater than a threshold of 0.5 I count the crop as being positive for Tuberculosis. I then sum up all of the positive crops for a slide and if more than half of the crops are counted positive I label the slide as positive. I run this for all 100 of the positive slides in the dataset and a subset of 100 of the 1000 negative slides in the dataset.

I've found the expected values for a well performing model differ based on whether the slide is positive or negative. A well performing model will be able to classify most or all 9000 of the crops from the negative slides as negative, however a well performing model should not classify all 9000 of the positive crops as positive. This is because on the positively labeled slides, only about 5–20% of the cropped fields actually appear to contain Tuberculosis. Therefore the percentage of positively labeled slides will be fairly low and the total number of positively predicted crops should be around 1000–2000 crops.

While this testing methodology is not perfect, it does give

fairly good insight into how well the model is able to sort the croppings by their likelihood, which is the ultimate goal of this project.

Crop Likelihood Viewer: To allow for manual validation of the model's performance a tool was constructed to visually sort a sputum slide by the likelihood of the presence of Tuberculosis Bacilli. This tool displays the various cropped view fields, sorted from most likely to contain TB to the least likely, as well as the total numbers and percentages of the positively and negatively predicted crops from the slide (See figure 3).

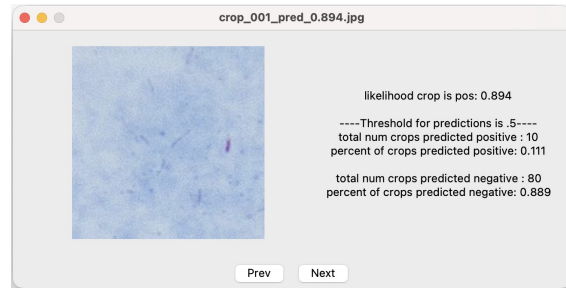


Figure 3. TB likelihood imaging tool.

4. Experiments

I used the Keras implementation as a baseline for my CNN, with a $(224 \times 224 \times 3)$ channel input schema, Binary Cross Entropy Loss, the Adam optimizer, the ReLU activation function for the convolutional layers, a sigmoid activation function for the output layer, and a learning rate of $1e-4$.

Further, the input images were split into a validation set and a training set via a 20/80 split respectively and normalized by dividing their pixel values by 255.

Training with only ODS3: I began my experiments working with my cropped modification of the ODS3 dataset to find a viable model architecture. I chose to be working with only ODS3 because it was the larger of the two training sets I had and it contains slides with less visual artifacts and staining distortion. I wanted to get my model working on a baseline input set, before I would look into attempting to improve generalization performance.

In my investigation, I found that the model appeared to overfit this dataset very early on in the training process. I attempted to combat this by selecting a simplistic architecture with a low number of filters and few convolutional layers, but then found that said architecture appeared to not be complex enough to converge well.

Ultimately, the best architecture I found for this dataset was 5 convolutional layers, interspersed with $(2,2)$ max pool

layers, starting from in the pattern of (64, 64, 128, 128, 256) convolutional filters.

This architecture achieved a validation accuracy of 0.9092 and a validation loss of 0.2901. By these results, the model can be said to be a fairly good classifier for the data, but it still needed to be tested with other datasets and on my testing set.

Augmenting with TBDB and constructing a Wellgen generalizable model: Once I included the TBDB data into my training and evaluation, I found that my model began to struggle to accurately classify with low loss. This is likely due to the fact that the TBDB images were at a higher resolution, so my (224×224) crops caused the Tuberculosis Bacilli to appear enlarged compared to the ODS3 images.

While this increased the training difficulty, I believe it is necessary to train on both of these sets as I am hoping to produce a generalizable model which can be used in many different lab environments where imaging setups might have different microscope magnification.

Further, the images in the Wellgen Medical subset, my testing set, are also captured from a third different lab environments, requiring even more generalization power from the model. To attempt to further increase my model's generalization I added an augmentation layer that runs every epoch and augments the training images, but not the testing images.

In this augmentation layer I added random flips and random rotations, zooms, brightness, and contrast operations with a factor of 0.4. These augmentations greatly decreased the likelihood my model overfit its training data, but also made training more difficult and added greater computational complexity.

Finally, I added batch normalization between my convolutional layers and my activation layers to further reduce the complexity of my weights. I also added dropout layers after the first and second max pooling layers with factors of 0.2 and 0.1 respectively and another final dropout layer with a factor of 0.5 after my global max pooling layer and before my final output activation. These dropout layers further reduced overfitting and the likelihood of having few convolutional filters capable of detecting important features.

With this setup I found the most performant model to be 5 convolutional layers, each with (3×3) kernels and increasing filter amounts of (32, 32, 64, 64, 128) and respective max pooling layers of size (2×2) after the first three convolutional layers which gradually decrease model complexity, but still allow for downstream feature extraction (See Figure 4).

Experimental Results: The increased model size and variability caused the convergence time to greatly increase, but

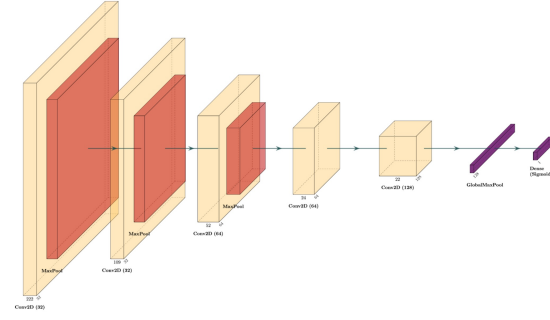


Figure 4. Final constructed model architecture.

the model also gained a much greater ability to generalize.

Training based on the architecture described above took about 12 hours of training time and 100 epochs to effectively converge. The augmentation layer applied to the training set caused deviations to appear in the observed loss, but overall the results are conducive to an effective convergence.

The minimum validation loss observed was 0.2097. This validation loss is low enough that it shows the model generalizes fairly well to unseen data, but there is certainly room for future improvement.

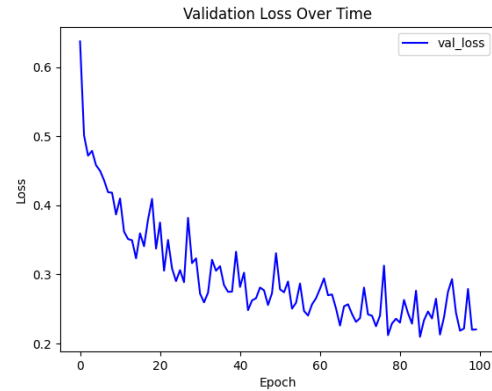


Figure 5. Graph of the validation loss from the 100 epoch run.

After 100 epochs, the training successfully achieved a 0.9198 validation accuracy. This accuracy score is quite high and is indicative that not only can the model be applied to unseen data, it can be used to accurately classify unseen data as well.

Similarly to the effect seen in the loss, the dropout layers caused spikes in the validation accuracy compared to the training accuracy. However, the spikes in the validation accuracy have a consistent deviation and still show gradual a

gradual increase over time as training continued (See Figure 6).

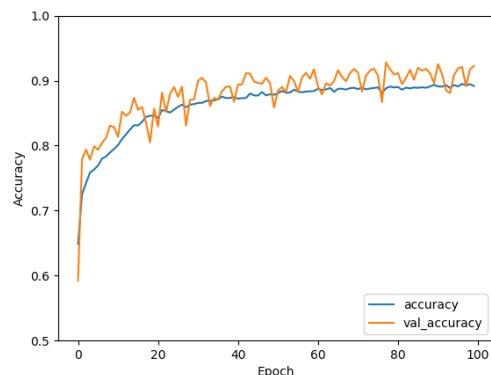


Figure 6. Graph of the training and validation accuracy from the 100 epoch run.

Finally, training for 100 epochs showed approximate convergence in the classifications of the crops from the positive and negative slides of the aforementioned Wellgen Medical subset used for testing.

As previously mentioned, the model was tested by taking 90 crops from both 100 positive slides and 100 negative slides for 9000 crops total for each class. Crops with an output likelihood of over a 0.5 probability threshold were counted as positive, while the ones under the 0.5 threshold were counted as negative.

The results from the classifications show the model was able to perform quite well in the classification task at later epochs. Because there are no tuberculosis found on negative slides it is expected a well performing model will classify around 9000 of the negative crops as negative. This was seen in the model's performance at later epochs. At the epoch with the best reported validation loss, the number of negative crops was 8836, this is very close to the target of 9000, thus providing evidence that the model is able to classify slides not containing TB quite well.

Accuracy interpretation is more complicated when it comes to the positive class. The positive slides tested tend to range from having about 5–20% of their croppings containing Tuberculosis Bacilli. Because of this, a well performing model should report somewhere between 1000 – 2000 slide crops as positive from the 9000 tested positive slides, and report the rest as negative. At the epoch with the best reported validation loss, the number of positive crops detected was 1134. This number of positive crops is within the expected range, thus providing further evidence for the model generalizing well (See Figure 7).

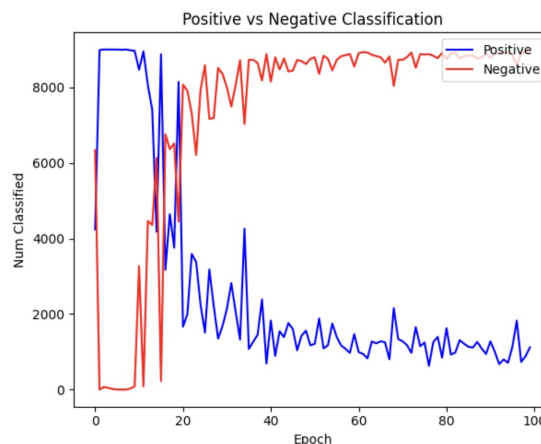


Figure 7. The number of positive and negative classifications over the 100 epoch run.

5. Discussion and Feature Analysis

Discussion: Detecting Tuberculosis via Ziehl-Neelsen microscopy is currently the cheapest and most accessible Tuberculosis diagnostic technique but is also very time-consuming and prone to false negatives. Because of this, a tool that can accurately flag areas with high potential for Tuberculosis presence and can then present them to be reviewed by a trained technician is highly useful.

Such a tool could be operated relatively inexpensively but greatly increase the rate of Tuberculosis detection and the speed of a diagnosis, cutting down the time technicians spend searching for bacterial presence on a slide. Not only would such a tool help a technician achieve better diagnostic results, speeding up the diagnostic time also means potentially freeing up time for other patients' sputum slides to be analyzed, increasing the overall rate of patient diagnostics.

Feature Analysis: The model appears to use features such as color, shape, and depth in its analysis. In images derived from the various convolutional layers it appears the filters are able to extract the three dimensionality of the bacilli in relation to the two dimensional slide they appear on (See Figure 8).

Further, I have noticed that the model performs better when there is a darker pink stain on the bacilli as opposed to bacilli stained with a less intense pink, leading me to believe that color is an important feature in detection as well.

Finally, it appears the shape of the bacilli is useful as the model is able to distinguish stained debris from actual bacilli with fairly good accuracy.

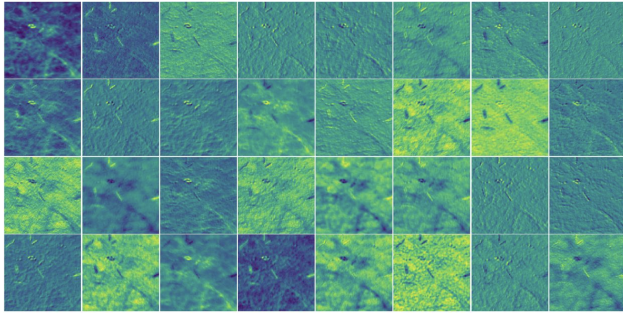


Figure 8. Visualization of the filters from the first convolutional layers, input image crop was predicted positive with a likelihood of 0.985.

6. Conclusion

The problem of detecting Tuberculosis is difficult because the places where it is most prevalent also lack adequate healthcare funding to do something about it. The most accessible diagnostic tool in the poorer regions where Tuberculosis is highly infectious is Ziehl-Neelsen stained slide microscopy. Microscopy is cheap and therefore accessible, but requires trained technicians to spend hours looking through microscope lenses to diagnose a single patient.

The classification system proposed in this article would be able to greatly speed up the rate at which a technician is able to classify Tuberculosis Bacilli by presenting areas of a viewfield to the technician where there is the greatest likelihood of Tuberculosis being present.

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